FraudLens AI

A suspicious transaction → an evidence-backed, reviewable fraud case in seconds

Theme: Intelligent Risk, Fraud & Financial Experiences (BFSI)

Built with: Gemini · Agent Development Kit · Cloud Run · Firestore

Prototype on synthetic data · all actions simulated

Team: [names] · AI Builder Cup 2026

The problem

- Banks process huge volumes of legitimate payments; fraud is a tiny fraction.
- **Sending every transaction to an LLM** is slow and expensive.
- **Fixed rules** flag customers whose normal behaviour simply differs, creating false positives.
- **Black-box scores** give analysts no evidence to act on.
- **Autonomous blocking by an LLM** is unsafe and unauditable.

Analysts need context, evidence and a controlled response, at scale.

Who we help

Persona	Today	With FraudLens
Fraud analyst	Alert queue + 5 screens to piece together context	One case workspace: evidence, timeline, network, AI summary
Risk / operations	Inconsistent responses	Deterministic, versioned risk tiers
Customer protection	Account takeover noticed late	Device, velocity and network signals combined
Auditors / ML team	Opaque decisions	Every claim tied to evidence IDs; full audit trail

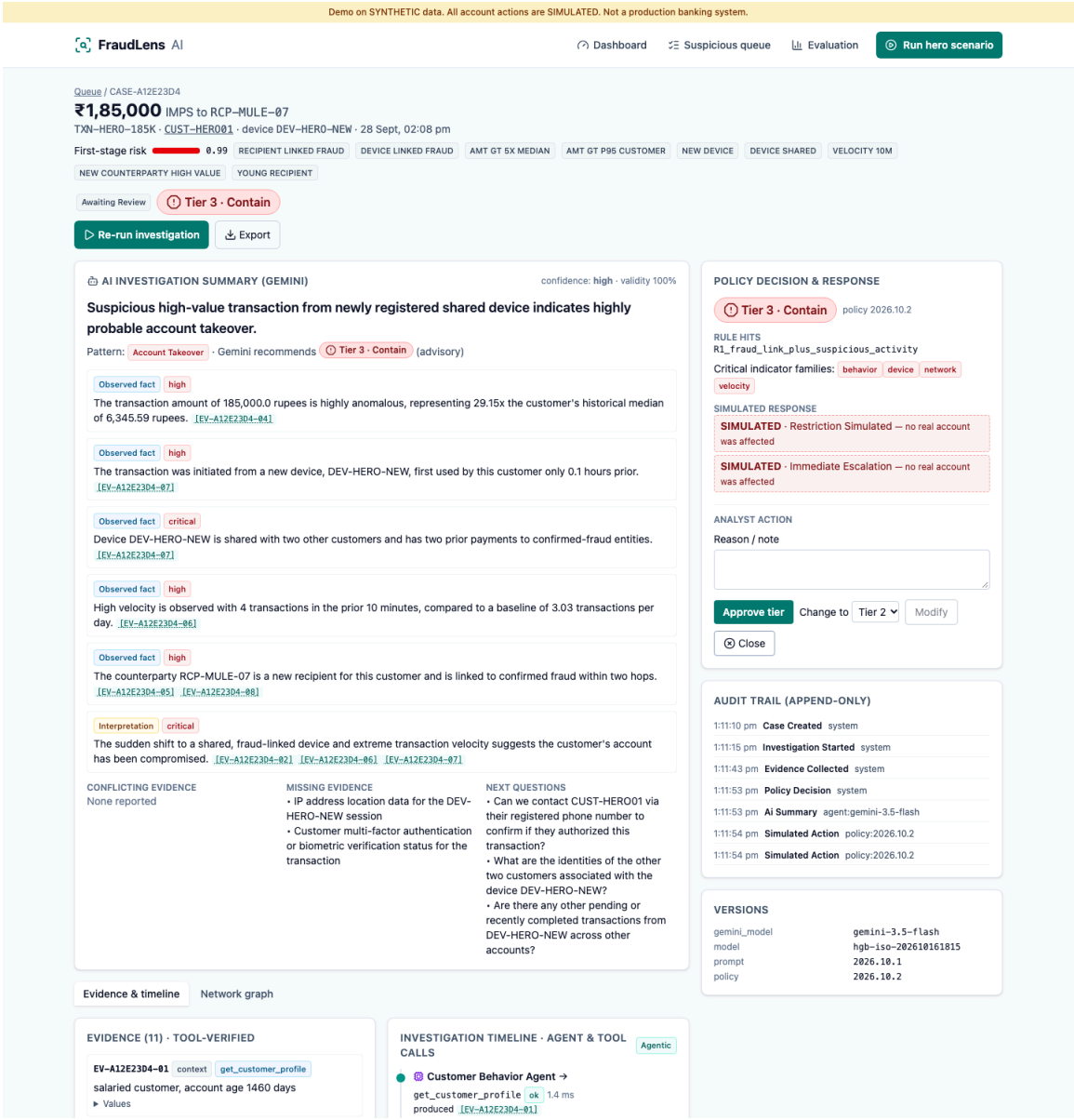
The solution: layered by design

ML handles scale → tools provide verified evidence → agents orchestrate → Gemini explains → policy controls the action

1. **Screen** every transaction with point-in-time features, rules and ML (no LLM).
2. **Investigate** only candidates with an ADK multi-agent team calling deterministic tools.
3. **Synthesize** with Gemini: structured JSON in which every finding cites evidence IDs.
4. **Decide** with a deterministic policy engine: Tier 1 Monitor / 2 Verify / 3 Contain.
5. **Respond** with a simulated action, analyst approval and an append-only audit trail.

Demo: the ₹1,85,000 transfer

- Customer normally makes 2–4 payments/day, median ≈ ₹6,000.
- A ₹1,85,000 IMPS transfer arrives from a device first seen minutes earlier, after 4 rapid payments.
- The recipient shares a device with confirmed-fraud mule accounts.
- Agents gather behaviour, velocity, device and network evidence.
- The policy engine applies **R1: fraud link + suspicious activity → Tier 3 Contain (SIMULATED)**.



How Gemini and agents make it possible

```
Orchestrator (Gemini, ADK)
  ├─ Customer Behavior Agent → deviation · history · limits
  ├─ Transaction Analysis Agent → velocity · device · counterparty · ML score
  └─ Fraud Network Agent → network paths · known fraud · related accounts
           ↓ every tool result becomes Evidence EV-…
Gemini synthesis → InvestigationSummary JSON (findings cite evidence IDs)
           ↓ citation validator drops unsupported claims
Policy engine (deterministic) → tier + allowed simulated actions
```

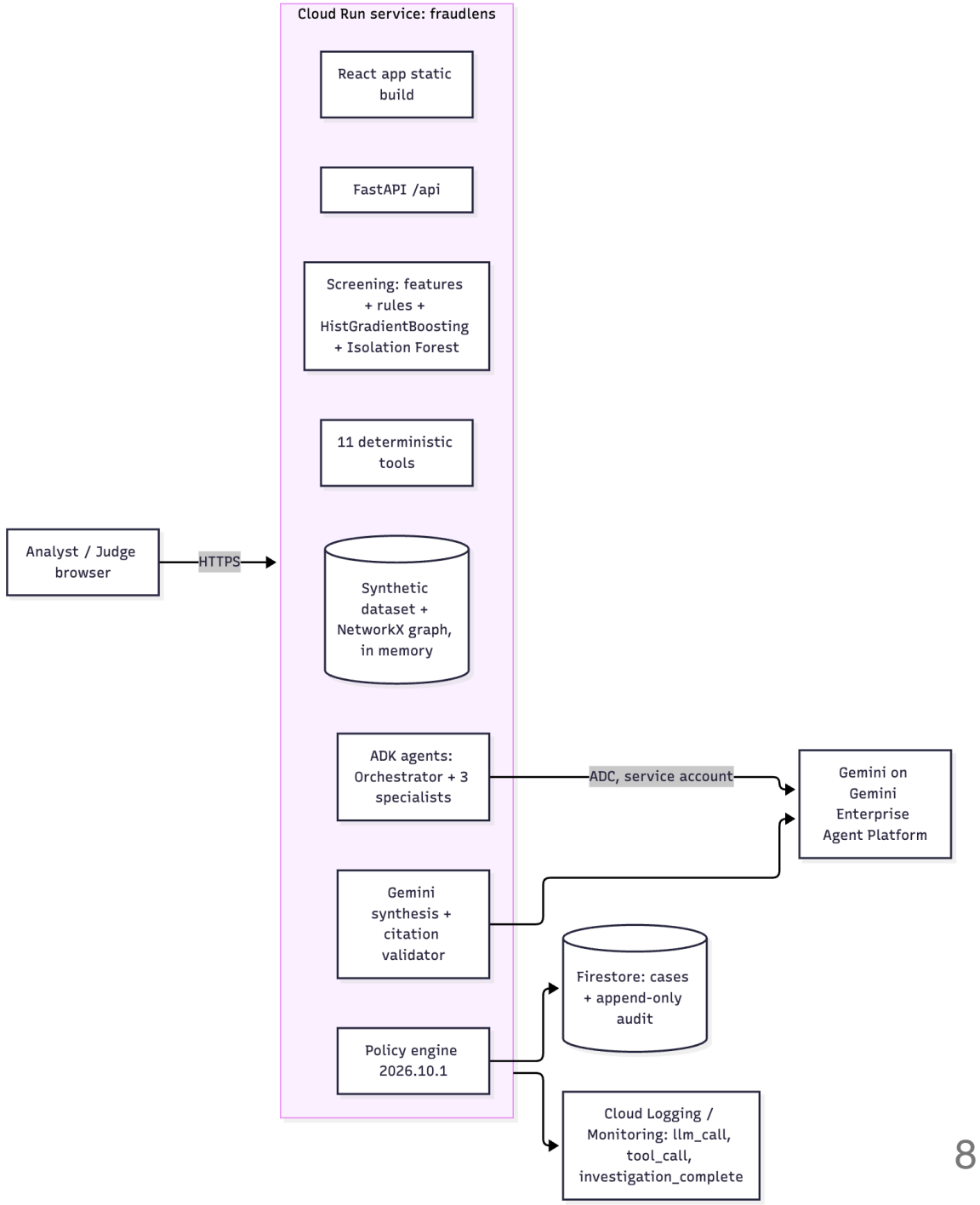
Callbacks block action tools, enforce a tool budget and validate IDs.

Grounded and safe by construction

- **Tools compute, Gemini reasons:** no LLM arithmetic, no invented values.
- **Evidence IDs on every claim;** unsupported findings are removed and the **evidence validity rate** is tracked.
- Observed facts vs interpretations; **missing and conflicting evidence** shown explicitly.
- **Gemini's tier is advisory;** the deterministic policy decides and disagreements are surfaced.
- Tool failure → evidence marked *unavailable*, never guessed. Gemini failure → manual review.
- Synthetic, pseudonymous data; bounded evidence package; append-only audit.

Architecture on Google Cloud

- **Cloud Run:** one container, UI + API
- **Gemini** via ADC, no API keys
- **ADK:** orchestrator + 3 specialists
- **Firestore:** cases + append-only audit
- **Cloud Logging:** LLM/tool calls, latency, tokens
- Least-privilege service account



Impact: how much better (held-out test split, 30,828 transactions)

Outcome	Rules only	FraudLens	Change
Frauds caught (of 214)	141	**182**	**+41 (+29%)**
Frauds missed	73	**32**	**–56%**
Ranking quality (PR-AUC)	0.6747	**0.8998**	**+33%**
LLM cost / 1,000 txns	$60.03 if every txn investigated	**$0.43**	**139× cheaper**

- **97.8%** of auto-escalated cases were real fraud (137 escalations)
- **100%** of Gemini findings grounded in real evidence, **0** invalid citations
- Account takeover 31→**62/62**, card testing 77→**98/98**

Trade-off we show, not hide: 39 false alarms vs 12, and mule/new-customer recall dropped. Next fix, already measured: rules **OR** ML catches **195 (91.1%)** at 79.9% precision.

Source: docs/metrics/impact.json (generated by evals/run_impact.py). No hand-entered numbers.

Measured results (synthetic held-out test split, 30,828 transactions)

Variant	Precision	Recall	PR-AUC	FP / 10k	Candidate rate
A · Rules only	0.9216	0.6589	0.6747	3.89	0.00496
B · ML + anomaly + rules	0.8235	0.8505	0.8998	12.65	0.00717
D · B + agentic Gemini investigation (real Gemini runs, policy 2026.10.2; 12 sampled test candidates + 2 golden scenarios)					

Gemini escalation rate	Tier-vs-label agreement	Evidence validity	Invalid citations	Latency p50 / p95	LLM cost / 1,000 txns
0.00717	0.8333	1.0	0	18.8 s / 23.2 s	$0.43037

C · B + Laya open decision model (zero-shot): routed 221 of 221 candidates to "investigate", so it matched B
(precision 0.8235, recall 0.8505) with 0.0 escalations avoided → **kept off the critical path** (BRD rule).

Source: docs/metrics/detection_metrics.json · benchmark.json · agentic_eval.json. No hand-entered numbers.

What's different

	Rules engine	ML score only	LLM chatbot	FraudLens
Scales to all transactions	✅	✅	❌	✅
Customer-specific baselines	❌	partly	❌	✅
Network / mule intelligence	❌	partly	❌	✅
Explains with evidence	codes	❌	unsupported text	✅ cited
Safe, auditable action	✅	❌	❌	✅ policy + human

Feasibility, scale and roadmap

Live today: screening, 11 tools, 4 agents, Gemini synthesis, policy engine, case workspace, audit, Cloud Run.

Scale path: BigQuery repository and Pub/Sub streaming · Cloud Run Jobs for batch · Spanner Graph · LLM cost tied
to candidate volume, not total volume.

Next 6 months: pilot with a bank fraud team on anonymised data · IAP/RBAC · threshold calibration on real
labels · analyst feedback loop · optional decision-model router benchmark.

Try it

Live demo: https://fraudlens-579191985047.us-central1.run.app
Video (≤ 3 min): [link]
Code: [GitHub repo]

Click **Run new investigation** → **₹1,85,000 account takeover** → watch the agents investigate → approve the Tier 3 case.

Synthetic data only. All account actions are simulated. Not a production banking system.